

Exercises

Level - 1

(Practice Questions Based on Fundamentals)

ABC OF COMPLEX NUMBER

- Find the value of $\log_3 \log_5 \log_3(243)$.
- Find the value of $\log_9(27) - \log_{27}(9)$.
- Find the value of $\log_{10} \tan 40^\circ + \log_{10} \tan 41^\circ + \log_{10} \tan 42^\circ + \dots + \log_{10} \tan 50^\circ$.
- Find the value of $\log_{\sqrt{3}} 300$, if $a = \log_{\sqrt{3}} 5$, $b = \log_{\sqrt{3}} 2$.
- Find the minimum value of
(i) $\log_b a + \log_a b$ (ii) $\log_b a + \log_c b + \log_a c$
- Prove that $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{43} n} = \frac{1}{\log_{43} n}$
- If $n = 1983!$, prove that
$$\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{1983} n} = 1$$
- Determine b satisfying
 $\log_a 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_a 10$
- If $\log_a bc = x$, find the value of $\log_b ab$.
- If $\log_{10} 2 = x$, find the value of $\log_{10} 5$.
- If $a = \log_4 5$ and $b = \log_5 6$, find $a = \log_3 2$.
- Find the value of $\log_{12} 54$, where $b = a = \log_{12} 24$.
- Find the value of $\frac{1}{\log_2 36} + \frac{1}{\log_3 36}$.
- Prove that $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > 2$.
- Simplify : $7 \cdot \log \frac{16}{15} + 5 \cdot \log \frac{25}{24} + 3 \cdot \log \frac{81}{80}$

16. If $a^2 + b^2 = 7ab$, prove that

$$\log \frac{1}{3}(a+b) = \frac{1}{2}(\log a + \log b)$$
17. If $a^2 + b^2 = 11ab$, prove that

$$\log \left(\frac{a-b}{3} \right) = \frac{1}{2}(\log a + \log b)$$
18. Prove that $\frac{\log_a n}{\log_{ab} n} = 1 + \log_a b$.
19. If $\log 25 = a$, $\log 225 = b$, prove that

$$\log \left(\left(\frac{1}{9} \right)^2 \right) + \log \left(\frac{1}{2250} \right) = 2a - 3b - 1$$
20. If a, b, c are in GP, prove that, $\log_a n, \log_b n, \log_c n$ are in HP.
21. If $\log_3 2, \log_3 (2^x - 5), \log_3 \left(2^x - \frac{7}{2} \right)$ are in AP, find the value of x .
22. If $y = a^{\frac{1}{1-\log_a x}}, z = a^{\frac{1}{1-\log_a y}}$, prove that, $x = a^{\frac{1}{1-\log_a z}}$.
23. If $x = \log_a c + \log_c a, y = \log_a c + \log_c a$ and $z = \log_b a + \log_a b$, find the minimum value of $x^2 + y^2 + z^2 - xyz$.
24. If $\frac{\log a}{b} \cdot \frac{\log b}{c} \cdot \frac{\log c}{a} = \frac{\log c}{a} \cdot \frac{\log a}{b}$, prove that $a^a \cdot b^b \cdot c^c = 1$.
25. Find the value of $81^{\frac{1}{\log_5 3}} + 27^{\frac{1}{\log_9 36}} + 3^{\frac{4}{\log_7 9}}$.
26. Prove that

$$\frac{1}{1 + \log_b a + \log_{bc} a} + \frac{1}{1 + \log_c a + \log_{ca} b} + \frac{1}{1 + \log_a b + \log_{ab} c} = 1$$
27. Find x , if $\log_2 x + \log_4 x + \log_8 x = 11$.
28. Find x , if $\log_2 x + \log_4 x + \log_8 x + \log_{16} x = \frac{25}{4}$.
29. If $\log_a x = \alpha, \log_b x = \beta, \log_c x = \gamma$ and $\log_d x = \delta$, find the value of $\log_{abcd} x$.
30. If $x = \log_a bc, y = \log_b ca$ and $z = \log_c ab$, find the value of $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$.
31. If $y = 2^{\frac{1}{\log_x 4}}$, find x .
32. If $N = 6^{\log_{10} 40} \cdot 5^{\log_{10} 36}$, find the value of $N + 10$.
33. Find the value of $(0.5)^{\log_{\sqrt{2}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \text{to } \infty \right)}$.
34. If $x = 2^{\log_{10} 3}$ and $y = 3^{\log_{10} 2}$, find a relation between x and y .
35. Find the value of

$$2^{\log_{10} 3 - \log_{10} 5} \times 3^{\log_{10} 5 - \log_{10} 2} \times 5^{\log_{10} 2 - \log_{10} 3}$$
36. If $a = \log_{30} 3$ and $b = \log_{30} 5$, find the value of $\log_{10} 8$.
37. If $a = \log_{12} 18, b = \log_{24} 54$, prove that $ab + 5(a - b) = 1$.

38. If $a = \log_7 12, b = \log_{12} 24$, find the value of $\log_{34} 168$.
39. If $a = \log_6 30, b = \log_{15} 24$, prove that

$$\log_{12} 60 = \left(\frac{2ab + 2a}{ab + b + 1} \right)$$
40. Find x , if $\log_7 (\log_5 (\sqrt{x+5} + \sqrt{x})) = 0$.
41. If $\log_2 x + \log_2 y \geq 6$, find the least value of $x + y$.
42. Solve for x and y :

$$4^{\log x} = 3^{\log y}, (3x)^{\log 3} = (4y)^{\log 4}$$
43. If $x^{18} = y^{21} = z^{28} = k$, prove that, $3, 3 \log_y x, 3 \log_z y, 7 \log_x z$ are in AP.
44. If $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > x$, find the value of x .
45. If $\log_{0.3} (x-1) < \log_{0.09} (x-1)$, find x .
46. If $\log_e \log_5 (\sqrt{2x-2} + 3) = 0$, find the value of x .
47. Find the least value of $2 \cdot \log_{10} x - \log_x (0.01)$ for $x > 1$.
48. Find x , if $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$
49. Find x , if $3^{4 \log_9 (x+1)} = 2^{2 \log_2 x} + 3$
50. If $a = \log_{24} 12, b = \log_{36} 24$ and $c = \log_{48} 36$, prove that

$$\left(\frac{abc + 1}{bc} \right) = 2$$
51. If $\log_{10} \left(\sin \left(x + \frac{\pi}{4} \right) \right) = \frac{1}{2}(\log_{10} 6 - 1)$, find the value of $\log_{10} \sin x + \log_{10} \cos x$.
52. If a, b, c are in GP, prove that

$$\frac{1}{1 + \log_e a}, \frac{1}{1 + \log_e b}, \frac{1}{1 + \log_e c}$$
 are in HP.
53. Find x , if $5^{\log_{10} x} = 50 \cdot x^{\log_{10} 5}$.
54. Find x , if $\log_5 [2 + \log (3 + x)] = 0$.

Find the number of real solutions

55. $\log_4 (x-1) = \log_2 (x-3)$
56. $\log_4 (x-2) = \log_2 (x-2)$
57. $\log_9 (x-1) = \log_3 (x-1)$
58. $\log_2 x + \log_2 (x+3) = 1/4$
59. $\log_4 (x^2 + x) - \log_4 (x+1) = 2$
60. $1 + 2 \log_{(x+2)} 5 = \log_5 (x+2)$
61. $\log_2 x + \log_4 (x+2) = 2$
62. $\log_{10} (x-1)^3 - 3 \log_{10} (x-3) = \log_{10} 8$

Solve for x :

63. $\log_5 (x^2 - 3x + 3) > 0$
64. $\log_7 [\log_5 (x^2 - 7x + 15)] > 0$
65. $\log_{(1/2)} [\log_5 (x^2 - 7x + 17)] > 0$
66. $\log_{(1/2)} (\log_5 (\log_2 (x^2 - 6x + 40))) > 0$
67. $\log_3 [\log_5 \log_2 (x^2 - 9x + 50)] > 0$

68. $\log_6 \left(\frac{x-2}{6-x} \right) > 0$
69. $\log_{(1/2)} x > \log_{(1/3)} x$
70. $\log_{0.5} (x^2 - 5x + 6) > -1$
71. $\log_8 (x^2 - 4x + 3) < 1$
72. $\log_{(1/4)} \left(\frac{35-x^2}{x} \right) \geq -\frac{1}{2}$
73. $\log_{(x^3+6)} (x^2-1) = \log_{(2x^2+5x)} (x^2-1)$
74. $\log (3x^2 + x - 3) = 3 \log (3x - 2)$
75. $\log_{(x^2-1)} (x^3+6) = \log_{(x^2-1)} (2x^2+5x)$
76. $\log_3 (x^2 - 3x - 5) = \log_3 (7 - 2x)$
77. $\log(\sqrt{x-1}) + \frac{1}{2} \log(2x+15) = 1$
78. $\log_{(3x+4)} (4x^2 + 4x + 1) + \log_{(2x+1)} (6x^2 + 11x + 4) = 4$
79. $5^{\log_{10} x} = 50 - x^{\log_{10} 5}$
80. $4^{\log_9 x} - 6.2^{\log_9 x} + 2^{\log_3 27} = 0$

Solve the in-equality wherever base is not given, take it as 10:

81. $(\log_2 x)^4 - \left(\log_{1/2} \left(\frac{x^5}{4} \right) \right)^2 - 20 \log_2 x + 148 < 0$
82. $(\log 100x)^2 + (\log 10x)^2 + \log x \leq 14$
83. $\log_{1/2} (x+1) > \log_2 (2-x)$
84. $\log_{1/5} (2x^2 + 5x + 1) < 0$
85. $\log_{1/2} x + \log_3 x > 1$
86. $\log_x \left(\frac{4x+5}{6-5x} \right) < -1$
87. $\log_{0.2} \left(\frac{x+2}{x} \right) \leq 1$
88. $\log_{10} (x^2 - 16) \leq \log_{10} (4x - 11)$

HINTS AND SOLUTIONS

LEVEL I

9. Given $\log_a(ab) = x$

$$\Rightarrow \log_a a + \log_a b = x$$

$$\Rightarrow 1 + \log_a b = x$$

$$\Rightarrow \log_a b = x - 1$$

Now, $\log_b(ab) = \log_b a + \log_b b$

$$= \log_b a + 1$$

$$= \frac{1}{x-1} + 1$$

$$= \frac{x}{x-1}$$

10. $\log_{10} 5 = \log_{10} \left(\frac{10}{2} \right)$

$$= \log_{10} 10 - \log_{10} 2 = (1 - x)$$

11. Now, $ab = \log_4 5 \cdot \log_5 6 = \log_4 6$

$$= \log_{2^2} (6)$$

$$= \frac{1}{2} \log_2 (6)$$

$$= \frac{1}{2} \log_2 (2 \times 3)$$

$$= \frac{1}{2} [\log_2 (2) + \log_2 3]$$

$$= \frac{1}{2} (1 + \log_2 3)$$

$$\Rightarrow 2ab = 1 + \log_2 3$$

$$\Rightarrow \log_2 3 = 2ab - 1$$

$$\Rightarrow \frac{1}{\log_2 3} = \frac{1}{2ab - 1}$$

$$\Rightarrow \log_3 2 = \frac{1}{(2ab - 1)}$$

13. We have

$$\frac{1}{\log_2 (36)} + \frac{1}{\log_3 (36)} = \log_{36} (2) + \log_{36} (3)$$

$$= \log_{36} (2 \times 3)$$

$$= \log_{36} (6)$$

$$= \log_{6^2} (6)$$

$$= 1/2$$

14. We have

$$\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} = \log_\pi 3 + \log_\pi 4$$

$$= \log_\pi (3 \times 4)$$

$$= \log_\pi (12) > \log_\pi (\pi^2) = 2$$

Hence, the value of x is 2.

21. Given $\log_3 2, \log_3 (2^x - 5), \log_3 \left(2^x - \frac{7}{2} \right)$ are in AP.

$$\Rightarrow 2 \log_3 (2^x - 5) = \log_3 2 + \log_3 \left(2^x - \frac{7}{2} \right)$$

$$\Rightarrow \log_3 (2^x - 5)^2 = \log_3 2 \cdot \left(2^x - \frac{7}{2} \right)$$

$$\Rightarrow (2^x - 5)^2 = 2 \times 2^x - 7$$

$$\Rightarrow (2^x)^2 - 12 \times 2^x + 32 = 0$$

$$\Rightarrow a^2 - 12a + 32 = 0 \text{ where } a = 2^x$$

$$\Rightarrow (a - 4)(a - 8) = 0$$

$$\Rightarrow a = 4, 8$$

$$\text{When } a = 4 \Rightarrow 2^x = 4 = 2^2 \Rightarrow x = 2$$

$$\text{when } a = 8 \Rightarrow 2^x = 8 = 2^3 \Rightarrow x = 3$$

But $x = 2$ does not satisfy the terms.

Hence, the solution of x is 3.

22. We have

$$y = a^{\frac{1}{1 - \log_a x}}, \quad z = a^{\frac{1}{1 - \log_a y}}$$

$$\Rightarrow \log_a y = \frac{1}{1 - \log_a x}, \quad \log_a z = \frac{1}{1 - \log_a y}$$

$$\Rightarrow \log_a y = \frac{1}{1 - \log_a x}, \quad \frac{1}{\log_a z} = 1 - \log_a y$$

$$\Rightarrow \log_a y = \frac{1}{1 - \log_a x}, \quad \log_a y = 1 - \frac{1}{\log_a z}$$

$$\text{Hence, } 1 - \frac{1}{\log_a z} = \frac{1}{1 - \log_a x}$$

$$\Rightarrow \frac{1}{\log_a z} = 1 - \frac{1}{1 - \log_a x}$$

$$\Rightarrow \frac{1}{\log_a z} = \frac{-\log_a x}{1 - \log_a x}$$

$$\Rightarrow \log_a z = \frac{1 - \log_a x}{-\log_a x} = 1 - \frac{1}{\log_a x}$$

$$\Rightarrow \frac{1}{\log_a x} = 1 - \log_a z$$

$$\Rightarrow \log_a x = \frac{1}{1 - \log_a z}$$

$$\Rightarrow x = a^{\frac{1}{1 - \log_a z}}$$

Hence, the result.

24. Let $\frac{\log a}{b} = \frac{\log b}{c} = \frac{\log c}{a} = k$

$$\Rightarrow \frac{a \log a}{a(b-c)} = \frac{b \log b}{b(c-a)} = \frac{c \log c}{c(a-b)} = k$$

$$\Rightarrow \frac{\log a^a}{a(b-c)} = \frac{\log b^b}{b(c-a)} = \frac{\log c^c}{c(a-b)} = k$$

$$\Rightarrow \log(a^a) + \log(b^b) + \log(c^c)$$

$$= k(ab - bc + bc - ba + ca - cb)$$

$$= 0$$

$$\Rightarrow \log(a^a b^b c^c) = 0$$

$$\Rightarrow (a^a b^b c^c) = c^0 = 1$$

27. We have,

$$\log_2 x + \log_4 x + \log_8 x = 11$$

$$\Rightarrow \log_2 x + \log_4 x + \log_8 x = 11$$

$$\Rightarrow \left(1 + \frac{1}{2} + \frac{1}{3}\right) \log_2 x = 11$$

$$\Rightarrow \left(\frac{11}{6}\right) \log_2 x = 11$$

$$\Rightarrow \log_2 x = 6$$

$$\Rightarrow x = 2^6 = 32$$

Hence, the value of x is 32.

30. Given

$$x = \log_a bc$$

$$\Rightarrow a_x = bc$$

$$\Rightarrow a \times a^x = abc$$

$$\Rightarrow a^{x+1} = abc$$

$$\Rightarrow x + 1 = \log_a (abc)$$

$$\Rightarrow \frac{1}{x+1} = \frac{1}{\log_a (abc)}$$

$$\text{Similarly, } \frac{1}{y+1} = \frac{1}{\log_b (abc)},$$

$$\text{and } \frac{1}{z+1} = \frac{1}{\log_c (abc)}$$

$$\text{Thus, } \frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$$

$$= \frac{1}{\log_a (abc)} + \frac{1}{\log_b (abc)} + \frac{1}{\log_c (abc)}$$

$$= \log_{abc} (a) + \log_{abc} (b) + \log_{abc} (c)$$

$$= \log_{abc} (abc)$$

$$= 1$$

31. We have $y = 2^{\frac{1}{\log_2 x^4}}$

$$y = 2^{\log_4 x} = 2^{\frac{1}{2} \log_2 x} = 2^{\log_2 (x)^{1/2}} = x^{1/2} = \sqrt{x}$$

36. As we know that

$$\log_{30} (30) = 1$$

$$\Rightarrow \log_{30} (2 \times 3 \times 5) = 1$$

$$\Rightarrow \log_{30} (2) + \log_{30} (3) + \log_{30} (5) = 1$$

$$\Rightarrow \log_{30} (2) + a + b = 1$$

$$\Rightarrow \log_{30} (2) = 1 - (a + b)$$

$$\text{Now, } \log_{30} (10) = \log_{30} (2 \times 5)$$

$$= \log_{30} (2) + \log_{30} (5)$$

$$= 1 - (a + b) + b = 1 - a$$

Thus,

$$\log_{10} 8 = \log_{10} (23) = 3 \log_{10} (2)$$

$$= 3 \times \frac{\log_{30} (2)}{\log_{30} (10)}$$

$$= \frac{3\{1 - (a + b)\}}{(1 - a)}$$

37. We have

$$ab = \log_{12} 18 \times \log_{24} 54$$

$$= \frac{\log 18}{\log 12} \times \frac{\log 54}{\log 24}$$

$$= \frac{\log (2 \times 3^2)}{\log (3 \times 2^2)} \times \frac{\log (2 \times 3^3)}{\log (3 \times 2^3)}$$

$$= \frac{\log 2 + 2 \log 3}{\log 3 + 2 \log 2} \times \frac{\log 2 + 3 \log 3}{\log 3 + 3 \log 2}$$

$$= \frac{1 + 2 \log_2 3}{\log_2 3 + 2} \times \frac{1 + 3 \log_2 3}{\log_2 3 + 3}$$

$$= \frac{1 + 2x}{x + 2} \times \frac{1 + 3x}{x + 3} \text{ where } x = \log_2 3$$

$$= \frac{1 + 5x + 6x^2}{x^2 + 5x + 6}$$

Also,

$$\begin{aligned}
 (a-b) &= \log_{12} 18 - \log_{24} 54 \\
 &= \frac{\log(2 \times 3^2)}{\log(3 \times 2^2)} - \frac{\log(2 \times 3^3)}{\log(3 \times 2^3)} \\
 &= \frac{\log 2 + 2 \log 3}{\log 3 + 2 \log 2} - \frac{\log 2 + 3 \log 3}{\log 3 + 3 \log 2} \\
 &= \frac{1 + 2 \log_2 3}{\log_2 3 + 2} - \frac{1 + 3 \log_2 3}{\log_2 3 + 3} \\
 &= \frac{1 + 2x}{x + 2} - \frac{1 + 3x}{x + 3} \text{ where } x = \log_2 3 \\
 &= \frac{(1-x^2)}{(2+x)(3+x)}
 \end{aligned}$$

Hence, the value of

$$\begin{aligned}
 5(a-b) + ab &= \frac{5(1-x^2)}{(2+x)(3+x)} + \frac{(1+5x+6x^2)}{(2+x)(3+x)} \\
 &= \frac{5(1-x^2) + (1+5x+6x^2)}{(2+x)(3+x)} \\
 &= \frac{x^2 + 5x + 6}{(2+x)(3+x)} \\
 &= \frac{(x+2)(x+3)}{(2+x)(3+x)} \\
 &= 1
 \end{aligned}$$

41. Given

$$\begin{aligned}
 \log_2 x + \log_2 y &\geq 6 \\
 \Rightarrow \log_2 (xy) &\geq 6 \\
 \Rightarrow xy &\geq 2^6 = 64
 \end{aligned}$$

$$\text{As we know that, } \frac{x+y}{2} \geq \sqrt{xy}$$

$$\Rightarrow \frac{x+y}{2} \geq \sqrt{64} = 8$$

$$\Rightarrow xy \geq 2.8 = 16$$

Hence, the least value of $x+y$ is 16.

43. Given,

$$x^{18} = y^{21} = z^{28}$$

$$\Rightarrow \log(x^{18}) = \log(y^{21}) = \log(z^{28})$$

$$\Rightarrow 18 \log x = 21 \log y = 28 \log z = k \text{ (say)}$$

$$\text{Now, } 3 \log_y x = 3 \cdot \frac{\log x}{\log y} = 3 \cdot \frac{21}{18} = \frac{7}{2}$$

$$3 \log_z y = 3 \cdot \frac{\log y}{\log z} = 3 \cdot \frac{28}{21} = 4$$

$$\text{and } 7 \log_x z = 7 \cdot \frac{\log z}{\log x} = 7 \cdot \frac{18}{28} = \frac{9}{2}$$

Thus, $3, 7/2, 4, 9/2$ are in AP.

Hence, the result.

48. Given equation is

$$4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$$

$$\Rightarrow 4^{\frac{1}{2} \log_3 3} + 9^{2 \log_2 2} = 10^{\log_x 83}$$

$$\Rightarrow 2 + 81 = 10^{\log_x 83}$$

$$\Rightarrow 83 = 83^{\log_x 10}$$

$$\Rightarrow \log_x 10 = 1$$

$$\Rightarrow x = 10$$

Hence, the solution is $x = 10$.

50. We have,

$$\begin{aligned}
 abc &= \log_{24} 12 \times \log_{36} 24 \times \log_{48} 12 \\
 &= \log_{48} 36 \times \log_{36} 24 \times \log_{24} 12 \\
 &= \log_{48} 12
 \end{aligned}$$

Now,

$$\begin{aligned}
 abc + 1 &= \log_{48} 12 + 1 = \log_{48} 12 + \log_{48} 48 \\
 &= \log_{48} (12 \times 48)
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } bc &= \log_{36} 24 \times \log_{48} 36 \\
 &= \log_{48} 36 \times \log_{36} 24 \\
 &= \log_{48} 24
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } \left(\frac{abc+1}{bc} \right) &= \frac{\log_{48} (12 \times 48)}{\log_{48} 24} \\
 &= \log_{24} (12 \times 48) \\
 &= \log_{24} (24 \times 48) \\
 &= \log_{24} (24^2) \\
 &= 2
 \end{aligned}$$

Hence, the result.

$$51. \text{ Given } \log_{10} \left(\sin \left(x + \frac{\pi}{4} \right) \right) = \frac{1}{2} (\log_{10} 6 - 1)$$

$$\Rightarrow 2 \log_{10} \left(\sin \left(x + \frac{\pi}{4} \right) \right) = (\log_{10} 6 - 1)$$

$$\Rightarrow 2 \log_{10} \left(\frac{1}{\sqrt{2}} (\sin x + \cos x) \right) = (\log_{10} 6 - 1)$$

$$\begin{aligned}
 \Rightarrow 2 \log_{10} \left(\frac{1}{\sqrt{2}} \right) + 2 \log_{10} (\sin x + \cos x) &= \log_{10} 6 - 1 \\
 &= \log_{10} 6 - 1
 \end{aligned}$$

$$\Rightarrow 2 \log_{10} (\sin x + \cos x) = \log_{10} 6 + 2 \log_{10} 2 - 1$$

$$\Rightarrow \log_{10} (\sin x + \cos x)^2 = \log_{10} \left(\frac{24}{10} \right)$$

$$\Rightarrow (\sin x + \cos x)^2 = \left(\frac{24}{10} \right)$$

$$\Rightarrow 1 + \sin 2x = \frac{24}{10}$$

$$\Rightarrow \sin 2x = \frac{24}{10} - 1 = \frac{14}{10}$$

$$\Rightarrow \sin x \cdot \cos x = \frac{7}{10}$$

$$\Rightarrow \log_{10} (\sin x \cos x) = \log_{10} \left(\frac{7}{10} \right)$$

$$\Rightarrow \log_{10} (\sin x) + \log_{10} (\cos x) = \log_{10} 7 - 1$$

52. Given a, b, c are in GP.

$$\therefore b^2 = ac$$